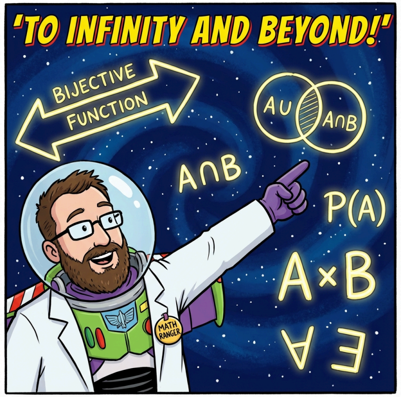
[illegible]

[illegible] :[illegible]

[illegible] .[illegible] :[illegible]

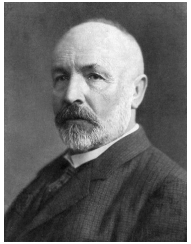
- [illegible]
- [illegible]


'TO INFINITY AND BEYOND!'
BIJECTIVE FUNCTION
A∪
A∩B
A∩B
P(A)
A×B
∀ ∃
MATH RANGER

- [illegible]
- [illegible]
- [illegible]

$\preccurlyeq$-[illegible] $\sim$ [illegible]

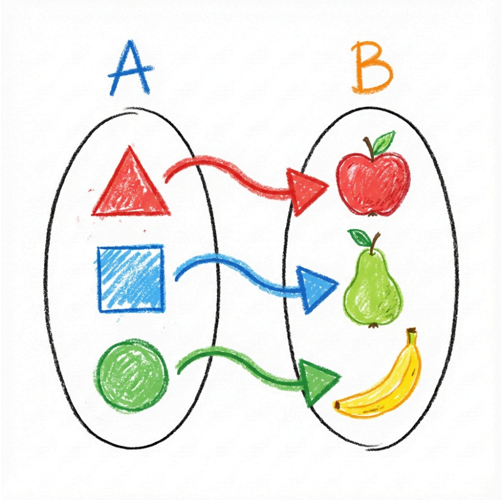
- [illegible] A [illegible]-B:
- $A \sim B$: [illegible] [illegible]-A [illegible]-B.
- $A \preccurlyeq B$: [illegible] [illegible]-A [illegible]-B.
- [illegible] [illegible]-$A \sim B \Rightarrow A \preccurlyeq B$.
- [illegible] $A \preccurlyeq B$ [illegible]-$B \preccurlyeq A$ [illegible] $A \sim B$.


[illegible]
1845-1918

[illegible]
1878-1956

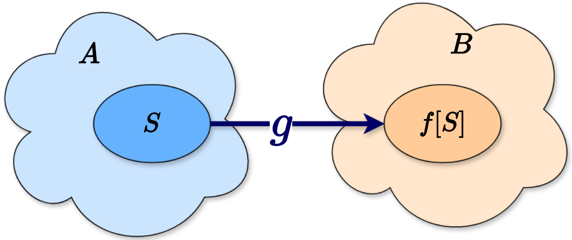
[illegible] $\sim$

- $\sim$ [illegible] $\mathcal{P}(U)$ [illegible] U.
- [illegible] ($A \sim A$):
  - [illegible] $id_A : A \to A$ [illegible] $id_A(a) = a$ [illegible]:
  - [illegible]: [illegible] $id_A(a_1) = id_A(a_2)$ [illegible] $a_1 = a_2$.
  - [illegible]: [illegible] $a \in A$, [illegible] $id_A(a) = a$.
- [illegible] ($A \sim B$ [illegible] $B \sim A$):
  - [illegible] $f : A \to B$ [illegible]. [illegible] $f^{-1} : B \to A$ [illegible].
  - [illegible]: [illegible] $f^{-1}(b_1) = f^{-1}(b_2) = a$, [illegible] f [illegible] $b_1 = f(a) = b_2$.
  - [illegible]: [illegible] $a \in A$, [illegible] $b = f(a)$. [illegible] $f^{-1}(b) = a$.
- [illegible] ($A \sim B$ -[illegible] $B \sim C$ [illegible] $A \sim C$):
  - [illegible] $f : A \to B$ -[illegible] $g : B \to C$. [illegible] $h = g \circ f : A \to C$.
  - [illegible]: $h(x) = h(y) \Rightarrow g(f(x)) = g(f(y)) \xrightarrow{g\ 1\text{-}1} f(x) = f(y) \xrightarrow{f\ 1\text{-}1} x = y$.
  - [illegible]: $c \in C$ [illegible] $b \in B$ [illegible] $g(b) = c$ -[illegible] ([illegible] g [illegible]). [illegible] $a \in A$ [illegible] $f(a) = b$ -[illegible] ([illegible] f [illegible]). [illegible] $h(a) = g(f(a)) = g(b) = c$.



[illegible] $A \sim B$ [illegible] $\mathcal{P}(A) \sim \mathcal{P}(B)$.

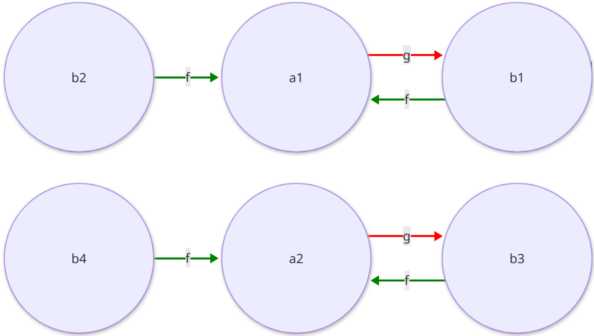
- [illegible] [illegible]-$f: A \to B$ [illegible] [illegible]-[illegible]-[illegible] [illegible].
- [illegible] [illegible] [illegible] $g: \mathcal{P}(A) \to \mathcal{P}(B)$ [illegible] [illegible] [illegible] $g(S) = f[S]$ [illegible] $S \subseteq A$.
- g [illegible] [illegible]-[illegible]-[illegible]:
  - [illegible] $g(S_1) = g(S_2)$ [illegible] $S_1, S_2 \subseteq A$.
  - [illegible] $f[S_1] = f[S_2]$.
  - [illegible] [illegible]-f [illegible] [illegible]-[illegible]-[illegible], [illegible] [illegible]-$S_1 = S_2$:
    - [illegible] $i \in \{1, 2\}$, [illegible] $x \in S_i$ [illegible] $f(x) \in f[S_i]$
    - [illegible] [illegible]-$f[S_{(2-i)+1}] = f[S_i]$, [illegible] [illegible]-$f(x) \in f[S_{(2-i)+1}]$
    - [illegible] [illegible]-f [illegible]-[illegible]-[illegible], [illegible] [illegible]-$x \in S_{(2-i)+1}$.
- g [illegible] [illegible]:
  - [illegible] $T \subseteq B$.
  - [illegible] $S = f^{-1}[T]$.
  - [illegible] $g(S) = T$.



[illegible] [illegible] [illegible] "[illegible]" [illegible] B [illegible] A, [illegible] $A \preccurlyeq B$

[illegible] [illegible] [illegible] "[illegible] [illegible] [illegible]" [illegible] [illegible] A [illegible] [illegible] ([illegible] [illegible] - [illegible] [illegible]).

[illegible], [illegible] $A = \emptyset$ [illegible]-$B = \{1\}$, [illegible] $A \preccurlyeq B$ ([illegible] [illegible]), [illegible] [illegible] [illegible] [illegible] [illegible] B-[illegible] A-[illegible].


b2
f
a1
g
f
b1
b4
f
a2
g
f
b3

- [illegible] $f: B \to A$ [illegible] [illegible] [illegible].
- [illegible] [illegible] [illegible] $g: A \to B$ [illegible]:
  - [illegible] [illegible] $a \in A$, [illegible] $b_a \in B$ [illegible] $f(b_a) = a$-[illegible] ([illegible] [illegible] [illegible] [illegible] f [illegible] [illegible] [illegible]
  - [illegible] $g(a) = b_a$.
- [illegible] g-[illegible] [illegible] [illegible]-[illegible]-[illegible]:
  - [illegible] $g(a_1) = g(a_2)$ [illegible] $a_1, a_2 \in A$.
  - [illegible] $b_{a_1} = b_{a_2}$.
  - [illegible] $f(b_{a_1}) = a_1$-[illegible] $f(b_{a_2}) = a_2$-[illegible], [illegible] $a_1 = a_2$-[illegible].
- [illegible] [illegible] g-[illegible] [illegible] [illegible]-[illegible]-[illegible], [illegible] $A \preccurlyeq B$.

[illegible] [illegible]: [illegible] $A \preccurlyeq B$ [illegible]-$A \neq \emptyset$, [illegible] [illegible] [illegible] "[illegible]" B-[illegible] A-[illegible]

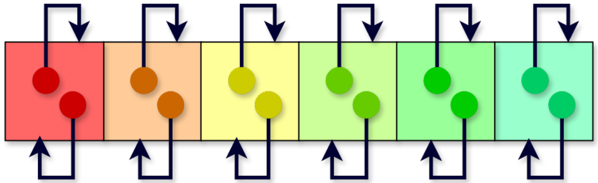
- [illegible] $g: A \to B$ [illegible]-[illegible]-[illegible], [illegible] [illegible] [illegible] $a_0 \in A$ ([illegible] $A \neq \emptyset$).
- [illegible] $f: B \to A$ [illegible]:

$$f(b) = \begin{cases} g^{-1}(b) & b \in g[A] \\ a_0 & b \notin g[A] \end{cases}$$

- [illegible] $b \in g[A]$, [illegible] $a \in A$ [illegible] $g(a) = b$-[illegible] ([illegible] g [illegible]"[illegible]), $g^{-1}(b)$-[illegible] [illegible] a-[illegible]. [illegible] f [illegible] [illegible].
- f [illegible] [illegible]:
- [illegible] $a \in A$. [illegible] $g(a) \in g[A]$, [illegible] $f(g(a)) = g^{-1}(g(a)) = a$.
- [illegible]: [illegible] $A \neq \emptyset$: [illegible] [illegible] [illegible] A-[illegible] B-[illegible] $\iff A \preccurlyeq B$.

[illegible] E [illegible] A, [illegible] $A/E \preccurlyeq A$

- [illegible] E [illegible] A.
- [illegible] $f\colon A \to A/E$ [illegible]: $f(a) = [a]$ [illegible] $a \in A$.
- [illegible] f-[illegible]:
  - [illegible] $[a] \in A/E$.
  - [illegible] $f(a) = [a]$ [illegible] $[a] \in \mathrm{image}(f)$.
- [illegible] $A/E \preccurlyeq A$.



[illegible] [illegible]: [illegible] $A \sim B$ [illegible] $C \sim D$ [illegible] $A \times C \sim B \times D$

[illegible] [illegible]: [illegible] $A \sim B$ [illegible] $C \sim D$ [illegible] $A \times C \sim B \times D$

- [illegible] $f\colon A \to B$ [illegible]-$g\colon C \to D$ [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible].
- [illegible] [illegible] $h\colon A \times C \to B \times D$ [illegible] [illegible]:

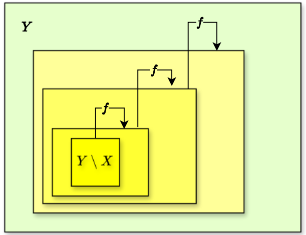
$$h(\langle a, c\rangle) = \langle f(a), g(c)\rangle$$

- [illegible] [illegible]-h [illegible] [illegible]-[illegible]-[illegible] [illegible]:

- [illegible]-[illegible]-[illegible]:
  - [illegible] $h(\langle a_1, c_1\rangle) = h(\langle a_2, c_2\rangle)$.
  - [illegible] $\langle f(a_1), g(c_1)\rangle = \langle f(a_2), g(c_2)\rangle$.
  - [illegible] $f(a_1) = f(a_2)$ [illegible] $g(c_1) = g(c_2)$.
  - [illegible] [illegible]-f [illegible]-g [illegible]"[illegible], [illegible] $a_1 = a_2$ [illegible]-$c_1 = c_2$.
  - [illegible] $\langle a_1, c_1\rangle = \langle a_2, c_2\rangle$.

- [illegible]:
  - [illegible] $\langle b, d\rangle \in B \times D$.
  - [illegible] [illegible]-f [illegible]-g [illegible], [illegible] $a \in A$ [illegible]-$c \in C$ [illegible] [illegible]-$f(a) = b$ [illegible]-$g(c) = d$.
  - [illegible] $h(\langle a, c\rangle) = \langle f(a), g(c)\rangle = \langle b, d\rangle$.

[illegible] [illegible]: [illegible] $X \subseteq Y$ [illegible] $f\colon Y \to X$ [illegible]-[illegible]-[illegible] [illegible] $X \sim Y$.



- [illegible] [illegible] E_n [illegible] [illegible]:

$$E_0 = Y \setminus X$$

$$E_{i+1} = f[E_i] = \{f(a) : a \in E_i\}$$

-

$$E = \bigcup\{E_i \mid i \in \mathbb{N}\}$$

- [illegible] [illegible] [illegible] $h\colon Y \to X$ [illegible]:

f(y) & \text{if } y \in E \\ y & \text{if } y \in Y \setminus E \end{cases}

- [illegible]: h [illegible] [illegible]-[illegible]-[illegible] [illegible].

[illegible]

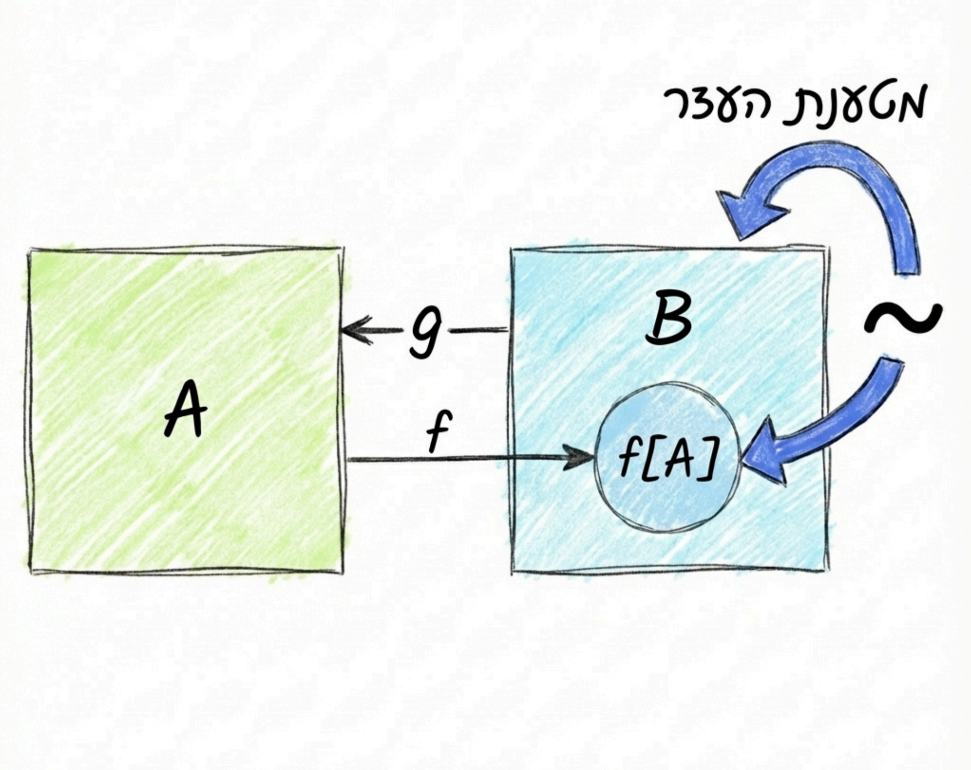
- h [illegible]-[illegible]-[illegible]:
- [illegible] $h(y_1) = h(y_2)$. [illegible] $y_1 = y_2$. [illegible]:

1. $y_1, y_2 \in E$: [illegible] $f(y_1) = h(y_1) = h(y_2) = f(y_2)$. [illegible] f-[illegible]-[illegible]-[illegible], [illegible] $y_1 = y_2$.
2. $y_1, y_2 \in Y \setminus E$: [illegible] $y_1 = h(y_1) = h(y_2) = y_2$. [illegible] $y_1 = y_2$.
3. $y_1 \in E$-[illegible] $y_2 \in Y \setminus E$: [illegible]. [illegible] $f(y_1) = h(y_1) = h(y_2) = y_2$. [illegible] $y_1 \in E$-[illegible], [illegible] $n \in \mathbb{N}$ [illegible]-$y_1 \in E_n$. [illegible] $y_2 = f(y_1) \in E_{n+1} \subseteq E$. [illegible] $y_2 \in E$, [illegible] $y_2 \in Y \setminus E$-[illegible].

- h [illegible]:
- [illegible] $x \in X$. [illegible] $x \in Y \setminus E$, [illegible] $h(x) = x$ [illegible] $x \in \mathrm{image}(h)$. [illegible] $x \in E$, [illegible] $x \in E_n$ [illegible] $n \in \mathbb{N}$. [illegible] $E_0 = Y \setminus X$-[illegible], $x \notin E_0$ [illegible] $n > 0$. [illegible] $x \in E_n$ [illegible] $n > 0$, [illegible] E_n, [illegible] $y \in E_{n-1}$ [illegible]-$x = f(y)$. [illegible] $y \in E_{n-1}$-[illegible], $y \in E$ [illegible] $h(y) = f(y) = x$ [illegible] $x \in \mathrm{image}(h)$.

[illegible] h-[illegible]-[illegible]-[illegible], [illegible] $X \sim Y$.

[illegible] [illegible] [illegible]: [illegible] $A \preccurlyeq B$ [illegible]-$B \preccurlyeq A$, [illegible] $A \sim B$.

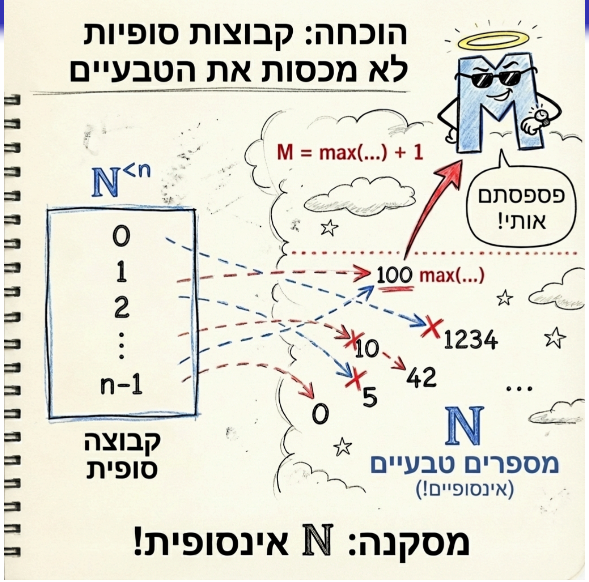

מטענת העזר
B
∼
g
A
f
f[A]

[illegible]: [illegible] $A \preccurlyeq B$ [illegible]-$B \preccurlyeq A$, [illegible] $A \sim B$.

[illegible]:

- [illegible] A [illegible]-B [illegible] $f : A \to B$ [illegible]-$g : B \to A$.
- $f \circ g$ [illegible].
- [illegible]-B [illegible]-$f[A]$
- [illegible]-$f[A] \sim B$.
- f [illegible] [illegible]-A [illegible]-$f[A]$, [illegible] $A \sim f[A]$.
- [illegible] $\sim$ [illegible]-$A \sim B$.

$\mathbb{N}$ [illegible]


הוכחה: קבוצות סופיות
לא מכסות את הטבעיים
M = max(...) + 1
פספסתם
אותי!
$\mathbb{N}^{<n}$
0
1
2
⋮
n-1
קבוצה
סופית
100 max(...)
10
1234
42
5
0
...
N
מספרים טבעיים
(אינסופיים!)
מסקנה: N אינסופית!

- [illegible] $\mathbb{N}$-[illegible].
- [illegible] $n \in \mathbb{N}$ [illegible]-$\mathbb{N}^{<n} \sim \mathbb{N}$.
- [illegible] $f : \mathbb{N}^{<n} \to \mathbb{N}$ [illegible].
- [illegible] $M = \max(\{f(i) \mid i < n\}) + 1$.
- [illegible], $M \in \mathbb{N}$ ([illegible]).
- [illegible], [illegible] $i < n$, [illegible] $f(i) < f(i) + 1 \leq M$, [illegible] $f(i) \neq M$.
- [illegible], $M \notin \text{image}(f)$.
- [illegible] f-[illegible].
- [illegible] $\mathbb{N}$ [illegible].

[illegible] $f: A \to A$ [illegible]-[illegible]-[illegible] [illegible] [illegible], [illegible] $\mathbb{N} \preccurlyeq A$

- [illegible] $a_0, a_1, \ldots$ [illegible]:
  - [illegible] $a_0 \in A \setminus \text{image}(f)$.
  - [illegible] $n \in \mathbb{N}$, [illegible] $a_{n+1} = f(a_n)$.
- [illegible] $g: \mathbb{N} \to A$ [illegible] $g(n) = a_n$.
- [illegible] g-[illegible] [illegible]-[illegible]-[illegible]: [illegible] g-[illegible] [illegible]". [illegible] $n < m$ [illegible] $g(n) = g(m)$.
  - [illegible] (n-[illegible] [illegible]) [illegible].

- [illegible] $n = 0$:
  - [illegible] $a_0 = g(0) = g(m) = f(a_{m-1})$.
  - [illegible], [illegible] $a_0 \notin \text{image}(f)$ [illegible].
- [illegible] $n > 0$:
  - [illegible] $m > 0$. [illegible] $f(a_{n-1}) = a_n = a_m = f(a_{m-1})$.
  - [illegible] f-[illegible]", [illegible] -[illegible] $g(n-1) = a_{n-1} = a_{m-1} = g(m-1)$.
  - [illegible] n.

- [illegible] g-[illegible] [illegible]-[illegible]-[illegible], [illegible] $\mathbb{N} \preccurlyeq A$.

[illegible]: [illegible] $\mathbb{Z}[x]$ [illegible]

- [illegible]-$\mathbb{Z}[x]$ [illegible].
- [illegible] $f:\mathbb{Z}[x]\to\mathbb{Z}[x]$ [illegible]:

$$f(P(x)) = x \cdot P(x)$$

- f [illegible]-[illegible]-[illegible]:
  - [illegible] $x \cdot P(x) = x \cdot Q(x)$, [illegible] $P(x) = Q(x)$ ([illegible]).
- f [illegible]:
  - [illegible] 1 [illegible] f, [illegible] $Q(x)$ [illegible], $Q(0) = 0 \cdot P(0) = 0$, [illegible] 1, [illegible]-0 [illegible] 1.
- [illegible]:
  - [illegible]-$\mathbb{Z}[x]$ [illegible].
  - [illegible], $\mathbb{N} \preccurlyeq \mathbb{Z}[x]$.
  - [illegible], $\mathbb{Z}[x]$ [illegible].

[illegible] [illegible] [illegible]

- [illegible] A [illegible] [illegible] [illegible] A [illegible] [illegible] $A \sim \mathbb{N}$.
- [illegible] $A \sim \mathbb{N}$ [illegible] [illegible]-A [illegible] [illegible] [illegible] [illegible] [illegible] $|A| = \aleph_0$ [illegible] [illegible] ([illegible]) [illegible] A [illegible] $\aleph_0$.
- [illegible]:
  - $2\mathbb{N} = \{0, 2, 4, 6, \dots\}$ [illegible] [illegible] [illegible] [illegible].
    - [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] $f : \mathbb{N} \to 2\mathbb{N}$ [illegible] $f(x) = 2x$.
    - [illegible] [illegible] [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible].
  - $\mathbb{Z}$ [illegible] [illegible] [illegible] [illegible]:
    - [illegible] [illegible] [illegible] $f : \mathbb{N} \to \mathbb{Z}$

      end{cases}\ { [illegible]}x \, \text ,\{ }frac{x}{2} & \text\- \\ { [illegible]}x \, \text ,\ {[illegible]-[illegible] }frac{x+1}{2} & \text\

    - [illegible] [illegible] [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible].

A [illegible] [illegible] [illegible] [illegible] [illegible] $A \preccurlyeq \mathbb{N}$

- (⇒) [illegible] A [illegible] [illegible] [illegible] [illegible] A [illegible] [illegible] $A \sim \mathbb{N}$.
  - [illegible] A [illegible], [illegible] [illegible] [illegible] [illegible] $f\colon A \to \mathbb{N}^{<n}$ [illegible] $n \in \mathbb{N}$ [illegible]. [illegible] [illegible] [illegible] A-[illegible] $\mathbb{N}$-[illegible] [illegible] $A \preccurlyeq \mathbb{N}$.
  - [illegible] $A \sim \mathbb{N}$ [illegible] [illegible] [illegible] [illegible] $A \preccurlyeq \mathbb{N}$.
- (⇐) [illegible] $A \preccurlyeq \mathbb{N}$, [illegible] [illegible] [illegible] $f\colon A \to \mathbb{N}$. [illegible] [illegible] [illegible] [illegible] [illegible]:

- [illegible] 1: $f[A]$ [illegible].
  - [illegible] [illegible], $f[A] \subseteq \mathbb{N}$ [illegible] [illegible]-[illegible] [illegible] [illegible] [illegible].
  - [illegible] $n \in \mathbb{N}$ [illegible] [illegible] [illegible] $g\colon f[A] \to \mathbb{N}^{<n}$.
  - [illegible] $g \circ f\colon A \to \mathbb{N}^{<n}$ [illegible] [illegible] [illegible].
  - [illegible] A [illegible], [illegible] [illegible] [illegible].

- [illegible] 2: $f[A]$ [illegible] [illegible].
  - [illegible] [illegible] $g\colon \mathbb{N} \to f[A]$ [illegible] [illegible]:

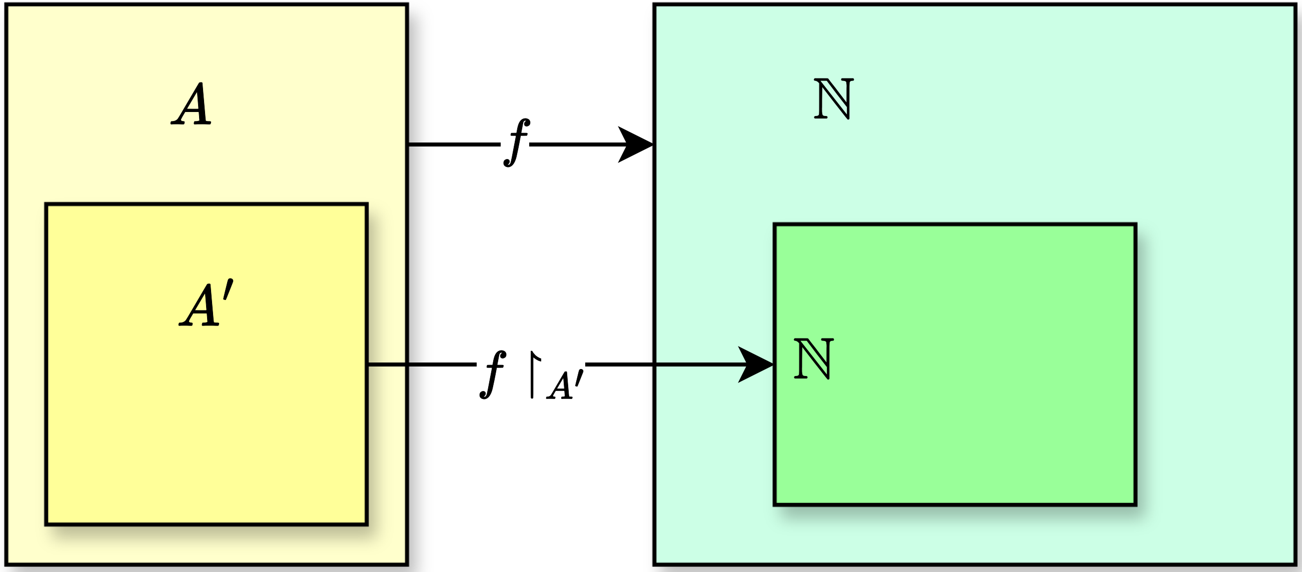
$$g(i) = \min(f[A] \setminus \{g(0), \ldots, g(i-1)\})$$

  - [illegible] [illegible] [illegible] $i \in \mathbb{N}$ [illegible] $f[A]$ [illegible] [illegible].
  - [illegible], [illegible] [illegible], g-[illegible] [illegible] [illegible] [illegible] [illegible].
  - [illegible] $f[A] \sim \mathbb{N}$. [illegible] f-[illegible] [illegible] [illegible] [illegible] A-[illegible] $f[A]$-[illegible], [illegible] $A \sim f[A]$. [illegible] [illegible], [illegible] $A \sim \mathbb{N}$.

- [illegible] g-[illegible] [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible]:
- g [illegible] [illegible]-[illegible]-[illegible]:
  - [illegible] $g(i) = g(j)$. [illegible] [illegible] [illegible]-$i = j$.
  - [illegible] [illegible] [illegible]-$i \neq j$. [illegible]"[illegible] [illegible] $i < j$.
  - [illegible] [illegible] g, [illegible] $g(j) = \min(f[A] \setminus \{g(0), \ldots, g(j-1)\})$.
  - [illegible] $g(j) \notin \{g(0), \ldots, g(j-1)\}$.
  - [illegible] [illegible], [illegible]-$i < j$, [illegible] $g(i) \in \{g(0), \ldots, g(j-1)\}$.
  - [illegible] $g(j) \neq g(i)$, [illegible] [illegible].
  - [illegible] [illegible]-$i = j$.

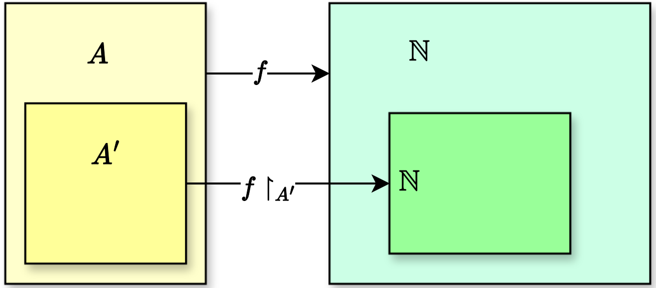
- g [illegible] [illegible]:
  - [illegible] $y \in f[A]$. [illegible] [illegible] $\{g(0), \ldots, g(y)\}$.
  - [illegible] [illegible] [illegible] $y + 1$ [illegible] [illegible] [illegible] ([illegible] g [illegible]"[illegible]).
  - [illegible] [illegible]-y [illegible] [illegible] g (y [illegible]). [illegible] [illegible] $k \leq y$ [illegible] $g(k) \neq y$.
  - [illegible] [illegible] [illegible], [illegible] $k \leq y$, $g(k) < y$ ([illegible] $y \in f[A]$ [illegible] [illegible], [illegible] [illegible] [illegible] [illegible]).
  - [illegible] $y + 1$ [illegible] [illegible] [illegible] [illegible] [illegible]-y. [illegible] [illegible] [illegible] [illegible] [illegible] y [illegible] [illegible] [illegible]-y: $0, \ldots, y - 1$.
  - [illegible] $y \in \{g(0), \ldots, g(y)\}$, [illegible] [illegible] $i \leq y$ [illegible]-$g(i) = y$.

[illegible] A' [illegible] ,$A' \subseteq A$-[illegible] [illegible] A [illegible]

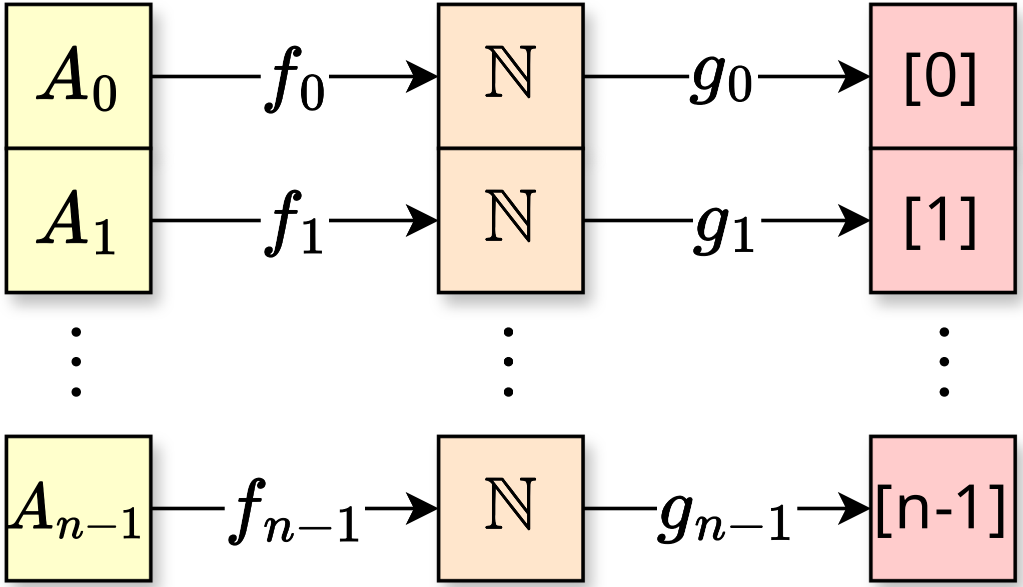

A
A'
f
$f\restriction_{A'}$
$\mathbb{N}$
$\mathbb{N}$

[illegible] A [illegible] [illegible] [illegible]-$A' \subseteq A$, [illegible] A' [illegible] [illegible]

- [illegible] [illegible]-A [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $A \preccurlyeq \mathbb{N}$.
- [illegible] $f: A \to \mathbb{N}$ [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible] [illegible] [illegible].
- [illegible] [illegible] [illegible] [illegible] [illegible] $f \restriction_{A'}: A' \to \mathbb{N}$.
- [illegible] [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible] $A' \preccurlyeq \mathbb{N}$ [illegible] [illegible]-A' [illegible] [illegible].

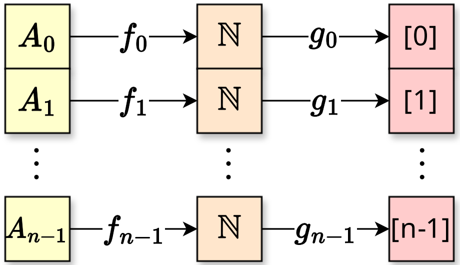

A
A'
f
$f \restriction_{A'}$
$\mathbb{N}$
$\mathbb{N}$

[illegible]


A_0
f_0
$\mathbb{N}$
g_0
[0]
A_1
f_1
$\mathbb{N}$
g_1
[1]
A_{n-1}
f_{n-1}
$\mathbb{N}$
g_{n-1}
[n-1]

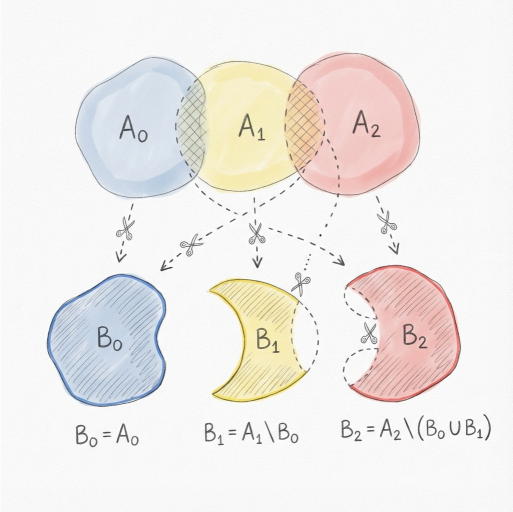
[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

- [illegible] $A_0, A_1, \ldots, A_{n-1}$ [illegible] [illegible] [illegible].
- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible]-$A_0, A_1, \ldots, A_{n-1}$ [illegible] [illegible] [illegible] [illegible].
- [illegible] [illegible] [illegible] [illegible] $A_0, A_1, \ldots, A_{n-1}$ [illegible] [illegible].
  - [illegible] $A_0, A_1, \ldots, A_{n-1}$ [illegible] [illegible] [illegible] [illegible] [illegible].
  - [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible] $f_i : A_i \to \mathbb{N}$.
  - [illegible] [illegible] $\mathbb{N}/R$ [illegible] R [illegible] $R = \{\langle i, j\rangle \in \mathbb{N} \times \mathbb{N} \mid i \equiv j \pmod{n}\}$.
  - [illegible] [illegible] [illegible] [illegible] $[0], [1], \ldots, [n-1]$. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]-[illegible] [illegible] [illegible] $\mathbb{N}$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]-$\mathbb{N} \sim [i]$ [illegible] $0 \leq i \leq n-1$.
  - [illegible] $g_i : \mathbb{N} \to [i]$ [illegible] [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible] [illegible] [illegible].
  - [illegible] [illegible] [illegible] $h : \bigcup_{i=0}^{n-1} A_i \to \mathbb{N}$ [illegible]-$h(a) = g_i(f_i(a))$ [illegible] $a \in A_i$.
  - [illegible] i, $g_i \circ f_i$ [illegible] [illegible]-[illegible]-[illegible] [illegible] [illegible]-A_i [illegible]-$[i]$
  - [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $\mathbb{N}$, h [illegible] [illegible]-[illegible]-[illegible] [illegible] [illegible].


A_0
f_0
$\mathbb{N}$
g_0
[0]
A_1
f_1
$\mathbb{N}$
g_1
[1]
A_{n-1}
f_{n-1}
$\mathbb{N}$
g_{n-1}
[n-1]

- [illegible] A_i [illegible]. [illegible] $B_0, B_1, \ldots, B_{n-1}$ [illegible]:

$$B_0 = A_0, \quad B_i = A_i \setminus \bigcup_{j=0}^{i-1} B_j$$


A₀
A₁
A₂
B₀
B₁
B₂
B₀ = A₀
B₁ = A₁ \ B₀
B₂ = A₂ \ (B₀ ∪ B₁)

1. [illegible]: $\bigcup_{i=0}^{n-1} A_i = \bigcup_{i=0}^{n-1} B_i$
   - [illegible] $B_i \subseteq A_i$-[illegible], [illegible] $\bigcup B_i \subseteq \bigcup A_i$.
   - [illegible], [illegible] $a \in \bigcup A_i$, [illegible] m [illegible] $a \in A_m$-[illegible].
   - [illegible] $a \notin A_k$ [illegible] $k < m$, [illegible] $a \notin B_k$.
   - [illegible] $a \in A_m \setminus \bigcup_{j=0}^{m-1} B_j = B_m$.
2. [illegible]: $B_i \cap B_j = \emptyset$ [illegible] $i \neq j$.
   - [illegible] $i > j$. [illegible], B_i [illegible] B_j-[illegible] ($B_j \subseteq \bigcup_{k=0}^{i-1} B_k$ [illegible]).

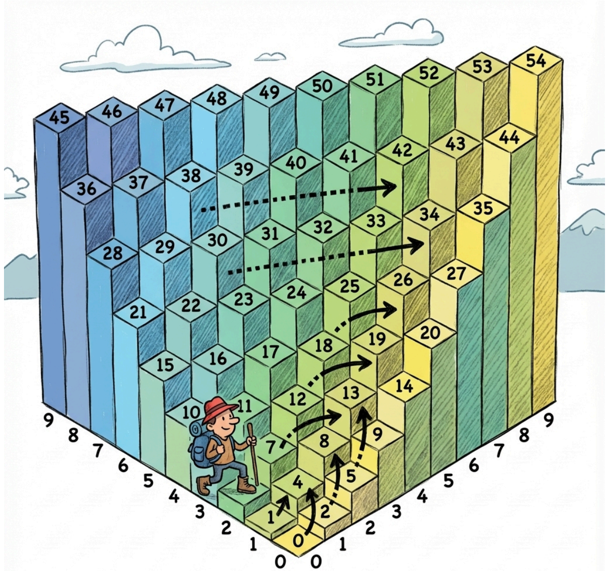
- [illegible], $\bigcup_{i=0}^{n-1} B_i$ [illegible] $\bigcup_{i=0}^{n-1} A_i$ [illegible].

$\mathbb{N} \times \mathbb{N}$ [illegible]

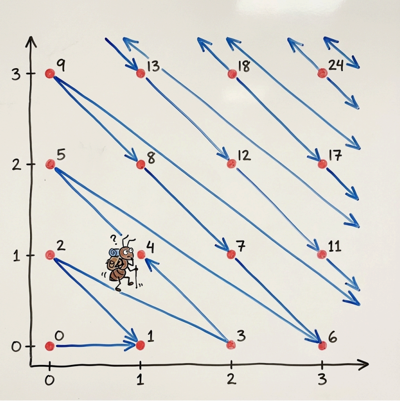
- [illegible] $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ [illegible]:

$$f(x,y) = \frac{(x+y)(x+y+1)}{2} + y$$

- [illegible] ([illegible]).
- [illegible], [illegible]-[illegible]:
  - $g(x,y) = 2^x 3^y$ [illegible] ([illegible]), [illegible] $\mathbb{N} \times \mathbb{N} \preccurlyeq \mathbb{N}$.
  - $x \mapsto \langle x, 0 \rangle$ [illegible], [illegible] $\mathbb{N} \preccurlyeq \mathbb{N} \times \mathbb{N}$.
  - [illegible]-[illegible], $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$.


0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54

[illegible]


0
1
2
3
4
5
6
7
8
9
11
12
13
17
18
24
?
0
1
2
3
0
1
2
3

[illegible]:

$$C(0) = (0,0)$$

[illegible]:

l, 0 \rangle & C(i)=\langle 0,y \rangle \\ \langle x-1, y+1\rangle & C(i)=\langle x,y \rangle, x\neq 0 \end{cases}\

1. [illegible] $\langle x,y \rangle$ [illegible] $\mathbb{N} \times \mathbb{N}$.
2. [illegible], [illegible]:
   - [illegible] d [illegible] $\langle x,y \rangle$ [illegible]-$x+y=d$.
   - [illegible] d [illegible]-$\langle d,0 \rangle$ [illegible] $\langle d-1,1\rangle, \langle d-2,2\rangle, \ldots, \langle 0,d \rangle$.
3. [illegible]-$\langle 0,0 \rangle$ [illegible], [illegible].
4. [illegible].

[illegible]: $\mathbb{Q}$ [illegible]

- [illegible] ([illegible]):

$\sim = \{\langle\langle a,b\rangle,\langle x,y\rangle\rangle \mid ay = xb\}$ [illegible] $\mathbb{Q} = (\mathbb{Z}\times(\mathbb{Z}\setminus\{0\}))/\sim$

- $\mathbb{Z}$ [illegible], [illegible] $\mathbb{Z}\setminus\{0\}$ ([illegible]-[illegible] $\mathbb{Z}$). [illegible] $\mathbb{Z}\sim\mathbb{N}$ [illegible]-$\mathbb{Z}\setminus\{0\}\sim\mathbb{N}$.
- [illegible] ($A\sim B$ [illegible]-$C\sim D \Rightarrow A\times C\sim B\times D$):

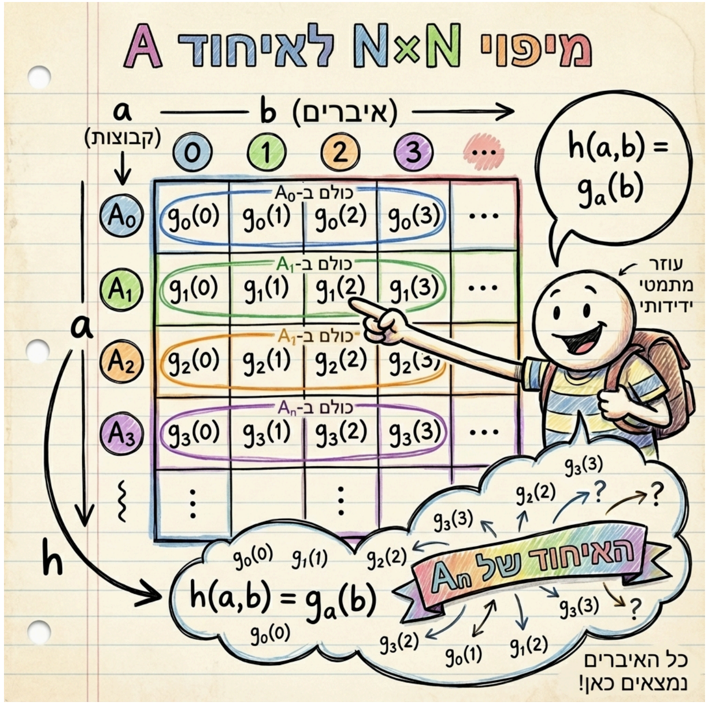
$$\mathbb{Z}\times(\mathbb{Z}\setminus\{0\})\sim\mathbb{N}\times\mathbb{N}\sim\mathbb{N}$$

- [illegible]-[illegible] [illegible] ([illegible] $\langle a,b\rangle\mapsto[\langle a,b\rangle]$), [illegible]

$$\mathbb{Q}\preccurlyeq\mathbb{Z}\times(\mathbb{Z}\setminus\{0\})\sim\mathbb{N}$$

- [illegible]-[illegible] [illegible]-[illegible], [illegible] $\mathbb{Q}$ [illegible].
- $\mathbb{Q}$ [illegible] ($n\mapsto\frac{n}{1}$ [illegible] $\mathbb{N}$-[illegible] $\mathbb{Q}$-[illegible]), [illegible] $|\mathbb{Q}|=\aleph_0$.

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]


מיפוי N×N לאיחוד A
a
(קבוצות)
b (איברים)
0
1
2
3
...
A_0
A_1
A_2
A_3
כולם ב-A_0
$g_0(0)$ $g_0(1)$ $g_0(2)$ $g_0(3)$ ...
כולם ב-A_1
$g_1(0)$ $g_1(1)$ $g_1(2)$ $g_1(3)$ ...
כולם ב-A_1
$g_2(0)$ $g_2(1)$ $g_2(2)$ $g_2(3)$
כולם ב-A_n
$g_3(0)$ $g_3(1)$ $g_3(2)$ $g_3(3)$ ...
⋮
a
h
h(a,b) = g_a(b)
עוזר מתמטי ידידותי
h(a,b) = g_a(b)
האיחוד של A_n
$g_0(0)$ $g_1(1)$ $g_2(2)$ $g_3(3)$ $g_2(2)$ $g_3(3)$? ?
$g_0(0)$ $g_3(2)$ $g_0(1)$ $g_1(2)$ $g_3(3)$?
כל האיברים נמצאים כאן!

- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $A = \{A_n \mid n \in \mathbb{N}\}$
- [illegible] [illegible] $h\colon \mathbb{N} \times \mathbb{N} \to \bigcup A$ [illegible] [illegible].
- [illegible] A_i [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] $g_i\colon \mathbb{N} \to A_i$ [illegible] [illegible].
- [illegible] [illegible] [illegible] h [illegible]: $h(\langle a, b\rangle) = g_a(b)$
- h [illegible] [illegible]:
  - [illegible] $a \in \bigcup A$.
  - [illegible] $a \in A_i$ [illegible] [illegible] [illegible] $i \in \mathbb{N}$.
  - [illegible], $a \in \operatorname{image}(g_i)$ [illegible] [illegible] $m \in \mathbb{N}$ [illegible] [illegible]-$a = g_i(m)$.
  - [illegible] [illegible]-$h(\langle i, m\rangle) = a$.
- [illegible] $\bigcup A \preccurlyeq \mathbb{N} \times \mathbb{N} \sim \mathbb{N}$.
- [illegible] $\bigcup A$ [illegible] [illegible].

[illegible] n [illegible]

- [illegible] $A_1, \ldots, A_n$ [illegible]. [illegible] n [illegible]:

$$A_1 \times \cdots \times A_n = \{\langle a_1, \ldots, a_n \rangle \mid a_1 \in A_1, \ldots, a_n \in A_n\}$$

- [illegible] n-[illegible]. [illegible] 2 [illegible]. [illegible] "[illegible]" [illegible].
- [illegible].
- [illegible] n [illegible]:
  - [illegible]: [illegible].
  - [illegible]: [illegible] $A_1 \times A_2 \times \cdots \times A_k$ [illegible].
    - [illegible] A_{k+1} [illegible] $(A_1 \times A_2 \times \cdots \times A_k) \times A_{k+1}$ [illegible].
    - [illegible] $f\colon A_1 \times A_2 \times \cdots \times A_k \times A_{k+1} \to (A_1 \times A_2 \times \cdots \times A_k) \times A_{k+1}$ [illegible] $f(\langle a_1, \ldots, a_k, a_{k+1} \rangle) = \langle \langle a_1, \ldots, a_k \rangle, a_{k+1} \rangle$ [illegible].

$\mathrm{FinSeq}(\mathbb{N})$ [illegible]

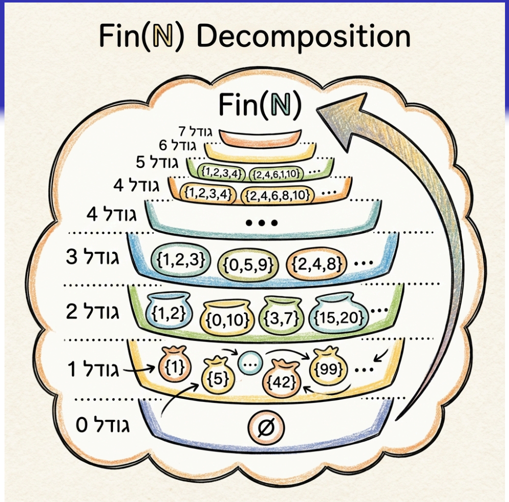
$$\mathrm{FinSeq}(\mathbb{N}) = \bigcup\{\mathbb{N}^m \mid m \in \mathbb{N}\} = \bigcup\{\{\langle a_1,\ldots,a_m\rangle \mid a_i \in \mathbb{N}\} \mid m \in \mathbb{N}\}$$

- [illegible].
- [illegible] $\mathrm{FinSeq}(\mathbb{N})$ [illegible].
- [illegible] $\mathbb{N}^{<n}$ [illegible]-$\mathbb{N}$ [illegible]:

$$\bigcup\{\mathbb{N}^{\mathbb{N}^{<n}} \mid n \in \mathbb{N}\}$$

- [illegible]-[illegible] [illegible]?
- [illegible] $\langle a_1,\ldots,a_m\rangle$ [illegible] $f : \mathbb{N}^{<m} \to \mathbb{N}$ [illegible]"[illegible] $f(i) = a_{i+1}$.

□□□□□□ □□ □□□□□□ □□□□ Fin(ℕ)


Fin(ℕ) Decomposition
Fin(ℕ)
גודל 7
גודל 6
גודל 5
{1,2,3,4}
{2,4,6,1,10}
גודל 4
{1,2,3,4}
{2,4,6,8,10}
גודל 4
גודל 3
{1,2,3}
{0,5,9}
{2,4,8}
גודל 2
{1,2}
{0,10}
{3,7}
{15,20}
גודל 1
{1}
{5}
{42}
{99}
גודל 0
∅

- □□□□□□ □-$\mathrm{Fin}(\mathbb{N})$ □□□□□ □□□□□□□ □□□□□□□□ □□□□□□□□□□□□□:

  1. $\emptyset \in \mathrm{Fin}(\mathbb{N})$.
  2. □□ $A \in \mathrm{Fin}(\mathbb{N})$ □-$x \in \mathbb{N}$, □□ $A \cup \{x\} \in \mathrm{Fin}(\mathbb{N})$.

- □□□□□□ □□ □□□□□□□□□ □□□□□□ $size\colon \mathrm{Fin}(\mathbb{N}) \to \mathbb{N}$ □□□□□□ □□□□□□□□□:
  - $size(\emptyset) = 0$
  - $$size(A \cup \{x\}) = \begin{cases} size(A) + 1 & x \notin A \\ size(A) & x \in A \end{cases}$$

- □□□ □□□□ □□□□□ □□ $\mathrm{Fin}(\mathbb{N})$ □□□□□□ □□ □□□□□ □□ □□□□□□□□ □□□□□ n: $\mathrm{Fin}(\mathbb{N}) = \bigcup\{\{A \in \mathrm{Fin}(\mathbb{N}) \mid size(A) = n\} \mid n \in \mathbb{N}\}$

- □□□□ $S_n = \{A \in \mathrm{Fin}(\mathbb{N}) \mid size(A) = n\}$. □□□□□□ □□□□□□□□□□□□ □□□ S_n □□□ □□ □□□□□:

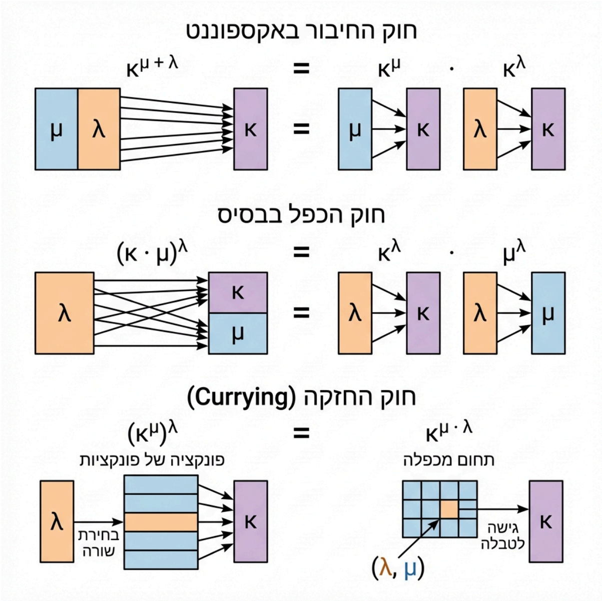
- □□□□□: $S_0 = \{\emptyset\}$ □□□ □□□□□ □□□□ □□ □□□□□.

- □□□: □□□□ □-S_n □□ □□□□□. □□ □□□□□ □-S_{n+1} □□□□□□□ □□ □□□ □□□□□ x □□□□□□□ □-S_n.
- □□□□□□ □□□□□□□□ □□ S_{n+1} □□□□□□□□ $F\colon S_n \times \mathbb{N} \to S_{n+1}$ □"□ $F(A, x) = A \cup \{x\}$.
- □□□□□ □-S_n □-$\mathbb{N}$ □□□□ □□□□□□, □□□□□□□ □□□□□ □□ □□□□□□, □□□□□ □□□□□□□ S_{n+1} □□ □□□□□□.

- □□□□□□: $\mathrm{Fin}(\mathbb{N})$ □□□ □□□□□□ □□ □□□□□□ □□ □□□□□□□ □□□□□ □□□□□□ (S_n), □□□□□ □□□ □□ □□□□□□.

[illegible] ([illegible])

- [illegible]:
  - [illegible]: $|A|+|B|=|A\uplus B|$ ([illegible]).
  - [illegible]: $|A|\cdot|B|=|A\times B|$ ([illegible]).
  - [illegible]: $|A|^{|B|}=|A^{B}|$ ([illegible] B-[illegible] A-[illegible]).
- [illegible]:
  - [illegible]: [illegible] ([illegible]).
  - [illegible]: [illegible]: $\kappa\cdot(\mu+\lambda)=\kappa\cdot\mu+\kappa\cdot\lambda$.
  - [illegible]:
    1. $\kappa^{\mu+\lambda}=\kappa^{\mu}\cdot\kappa^{\lambda}$
    2. $(\kappa\cdot\mu)^{\lambda}=\kappa^{\lambda}\cdot\mu^{\lambda}$
    3. $(\kappa^{\mu})^{\lambda}=\kappa^{\mu\cdot\lambda}$


חוק החיבור באקספוננט
κ^{μ + λ} = κ^μ · κ^λ
μ λ κ = μ κ λ κ
חוק הכפל בבסיס
(κ · μ)^λ = κ^λ · μ^λ
λ κ μ = λ κ λ μ
חוק החזקה (Currying)
(κ^μ)^λ = κ^{μ · λ}
פונקציה של פונקציות
תחום מכפלה
λ
בחירת שורה
κ
גישה לטבלה
κ
(λ, μ)

[illegible] $\{0,1\}^{\mathbb{N}}$

$$
\begin{array}{l}
f(1) = 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ \ldots \\
f(2) = 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ \ldots \\
f(3) = 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ \ldots \\
f(4) = 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ \ldots \\
f(5) = 1\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ \ldots \\
f(6) = 0\ 0\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 1\ 0\ \ldots \\
f(7) = 1\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 1\ 0\ 0\ \ldots \\
f(8) = 0\ 0\ 1\ 1\ 0\ 0\ 1\ 1\ 0\ 0\ 1\ \ldots \\
f(9) = 1\ 1\ 0\ 0\ 1\ 1\ 0\ 0\ 1\ 1\ 0\ \ldots \\
f(10) = 1\ 1\ 0\ 1\ 1\ 1\ 0\ 0\ 1\ 0\ 1\ \ldots \\
f(11) = 1\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ \ldots \\
\ \vdots \qquad \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \vdots\ \ddots
\end{array}
$$

$$
g = 1\ 0\ 1\ 1\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ \ldots
$$

- [illegible] -$\{0,1\}^{\mathbb{N}}$ [illegible].
- [illegible] $f:\mathbb{N}\to\{0,1\}^{\mathbb{N}}$.
- [illegible] $g:\mathbb{N}\to\{0,1\}$ [illegible] -$g(n) = 1 - f(n)(n)$
- [illegible] -f [illegible], [illegible] $m\in\mathbb{N}$ [illegible] -g $f(m) =$.
- [illegible]: $g(m) = 1 - f(m)(m) = 1 - g(m)$
- [illegible], [illegible] -$\{0,1\}^{\mathbb{N}}$ [illegible].
- [illegible] -$\{0,1\}^{\mathbb{N}}$ [illegible].

[illegible]-$\mathbb{R}$ [illegible]

- [illegible] $\mathbb{R}$ [illegible], [illegible] $[0,1]$ [illegible].
- [illegible] $[0,1]$ [illegible] 2 [illegible].
  - [illegible], $1/2 = 0.1\overline{0}$, $1/3 = 0.\overline{01}$, $3/4 = 0.11\overline{0}$.
- [illegible] r [illegible]:
  - [illegible]-$f(n) = r_n$ [illegible] r_n [illegible] 2 [illegible].
  - [illegible]-n-[illegible] r [illegible]-n-[illegible] r_n.
- [illegible]-r [illegible] r_n [illegible]-n-[illegible], r [illegible] f.
- [illegible], [illegible]-$\mathbb{R}$ [illegible].

[illegible]?

[illegible] - [illegible] $\mathbb{R}$

- [illegible]-$\mathbb{R}$ [illegible].
- [illegible] 0 [illegible]-1 [illegible]:

$$\left\{\sum_{i=1}^{\infty} d_i 10^{-i} \colon \forall i \in \mathbb{N} (d_i \in \{0,1\})\right\} \sim \mathbb{N}$$

- [illegible] 0 [illegible]-1.
- [illegible] $\{0,1\}^{\mathbb{N}}$:

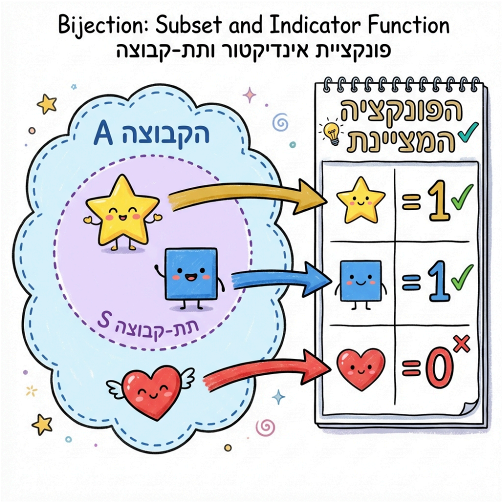
$$f(\sum_{i=1}^{\infty} d_i 10^{-i})(i) = d_i$$

- [illegible]-$\{0,1\}^{\mathbb{N}}$ [illegible].

$\mathcal{P}(A) \sim \{0,1\}^A$

- [illegible] $\mathcal{P}(A)$ [illegible]-A [illegible]-$\{0,1\}$.
- [illegible] $S \subseteq A$ [illegible] (Indicator Function), $\chi_S: A \to \{0,1\}$, [illegible]:

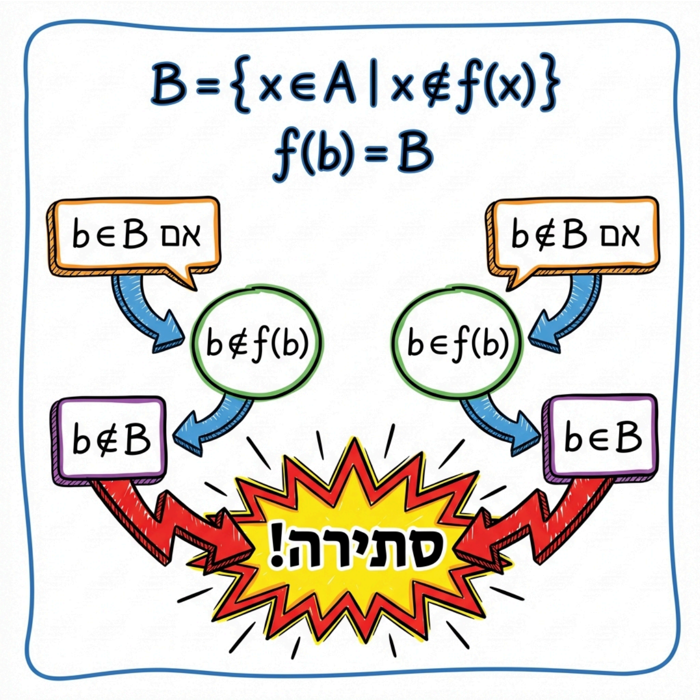
x \in S \\ 0 & x \notin S \end{cases} & 1


Bijection: Subset and Indicator Function
פונקציית אינדיקטור ותת-קבוצה
הקבוצה A
תת-קבוצה S
הפונקציה המציינת
=1
=1
=0

- [illegible] 1 [illegible]-S, [illegible]-0 [illegible]-S.
- [illegible], $\mathcal{P}(A) \sim \{0,1\}^A$.

[illegible] [illegible]: $\mathcal{P}(A) \not\sim A$

- [illegible] [illegible]-$\mathcal{P}(A) \sim A$
- [illegible] [illegible] [illegible]-[illegible]-[illegible] [illegible] $f: A \to \mathcal{P}(A)$.
- [illegible] [illegible] $B \subseteq A$ [illegible] [illegible] $B = \{a \in A \mid a \notin f(a)\}$.
- [illegible] [illegible]-f [illegible] [illegible], [illegible] $b \in A$ [illegible] [illegible]-$f(b) = B$.
- [illegible] [illegible] $b \in B$:
  - [illegible] $b \in B$, [illegible] [illegible] [illegible] B, $b \notin f(b)$, [illegible] $b \notin B$.
  - [illegible] $b \notin B$, [illegible] [illegible] [illegible] B, $b \in f(b)$, [illegible] $b \in B$.
- [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible]-$\mathcal{P}(A) \sim A$ [illegible] [illegible].


B = {x∈A | x∉f(x)}
f(b) = B
אם b∈B
אם b∉B
b∉f(b)
b∈f(b)
b∉B
b∈B
סתירה!

[illegible] [illegible] [illegible]: $A \prec \mathcal{P}(A)$

- [illegible] $a \mapsto \{a\}$ [illegible]-A [illegible]-$\mathcal{P}(A)$ [illegible] [illegible]"[illegible] ([illegible] $\{a\} = \{b\}$ [illegible] $a = b$), [illegible] $A \preccurlyeq \mathcal{P}(A)$.
- [illegible] [illegible] [illegible] [illegible]: $A \preccurlyeq \mathcal{P}(A)$ [illegible] $A \not\sim \mathcal{P}(A)$. [illegible] [illegible] $A \prec \mathcal{P}(A)$.
- [illegible] $A \not\succcurlyeq \mathcal{P}(A)$: [illegible], [illegible] [illegible]-[illegible], [illegible] [illegible] $A \sim \mathcal{P}(A)$.
- [illegible]: $\mathcal{P}(\mathbb{N})$ [illegible] [illegible] [illegible].
  - [illegible] $\mathcal{P}(\mathbb{N})$ [illegible] [illegible] [illegible], [illegible] [illegible] $\mathbb{N} \preccurlyeq \mathcal{P}(\mathbb{N})$, [illegible] [illegible] [illegible].
  - ([illegible] [illegible] [illegible] $\mathcal{P}(\mathbb{N}) \sim \{0,1\}^{\mathbb{N}}$ [illegible] [illegible] [illegible] [illegible].)
- [illegible] "[illegible] [illegible] [illegible]":

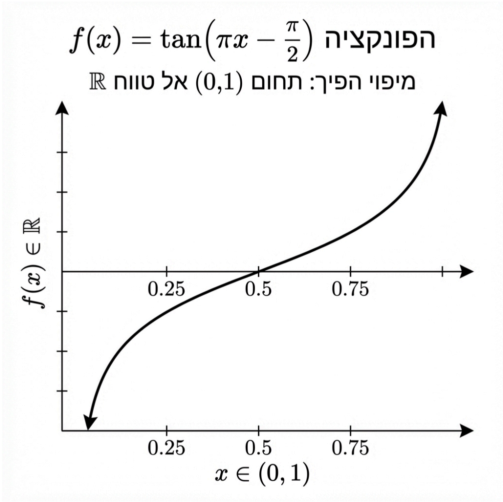
$$\mathbb{N} \prec \mathcal{P}(\mathbb{N}) \prec \mathcal{P}(\mathcal{P}(\mathbb{N})) \prec \cdots$$

[illegible] $(0,1)$ [illegible] $\mathbb{R}$

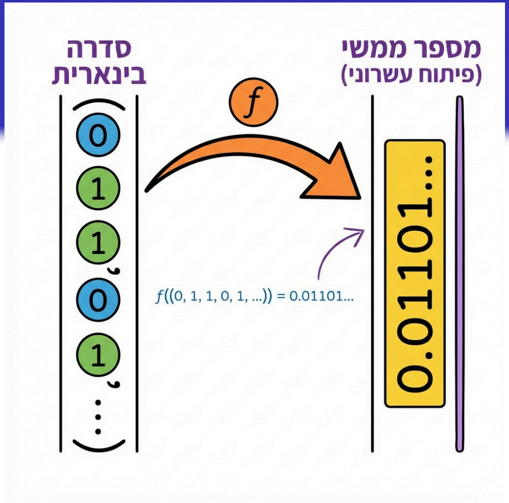
- [illegible] $\mathbb{R} \sim (0,1)$ [illegible].
- [illegible] $f:(0,1)\to\mathbb{R}$ [illegible]:

$$f(x)=\tan\left(\pi x-\frac{\pi}{2}\right)$$

- [illegible]:
  - [illegible] $(0,1)$.
  - [illegible] $x\to 0^+$, $f(x)\to-\infty$.
  - [illegible] $x\to 1^-$, $f(x)\to+\infty$.
  - [illegible] $(-\infty,\infty)$ [illegible].
- [illegible] f-[illegible], [illegible] $\mathbb{R}\sim(0,1)$.



$\{0,1\}^{\mathbb{N}} \sim \mathbb{R}$


סדרה בינארית
מספר ממשי (פיתוח עשרוני)
f
(0, 1, 1, 0, 1, ...)
$f((0, 1, 1, 0, 1, ...)) = 0.01101...$
0.01101...

- [illegible]: $\{0,1\}^{\mathbb{N}} \preccurlyeq \mathbb{R}$
- [illegible] $f : \{0,1\}^{\mathbb{N}} \to \mathbb{R}$ [illegible]:
  - [illegible] $a = (a_0, a_1, a_2, \ldots) \in \{0,1\}^{\mathbb{N}}$.
  - [illegible] $f(a)$ [illegible] a:

$$f(a) = 0.a_0a_1a_2\ldots = \sum_{n=0}^{\infty} \frac{a_n}{10^{n+1}}$$

- [illegible] f [illegible]-[illegible]-[illegible]:
  - [illegible] $f(a)$ [illegible] 0 [illegible]-1.
  - [illegible], [illegible] 9-[illegible] 0-[illegible]. [illegible] 9, [illegible].
  - [illegible], [illegible] $a \neq b$, [illegible] $f(a)$ [illegible]-$f(b)$ [illegible], [illegible] $f(a) \neq f(b)$.

([illegible]) $\{0,1\}^{\mathbb{N}} \sim \mathbb{R}$

- [illegible]: $\mathbb{R} \preccurlyeq \{0,1\}^{\mathbb{N}}$
- [illegible] $(0,1)$-[illegible] $\sim \mathbb{R}$. [illegible] $\{0,1\}^{\mathbb{N}}$-[illegible] $\succcurlyeq (0,1)$.
- [illegible] $x \in (0,1)$ [illegible] $b_n \in \{0,1\}$ ([illegible] 1-[illegible]):

$$x = 0.b_1b_2b_3\ldots_2 = \sum_{n=1}^{\infty} \frac{b_n}{2^n}$$

- [illegible] $g\colon (0,1) \to \{0,1\}^{\mathbb{N}}$ [illegible] x [illegible]: $g(x) = (b_1, b_2, b_3, \ldots)$
- [illegible] g [illegible]-[illegible]-[illegible]:
  - [illegible] $x \neq y$, [illegible] ([illegible] 1-[illegible]).
  - [illegible], [illegible] $g(x) \neq g(y)$.
- [illegible]: [illegible]-[illegible]: $\{0,1\}^{\mathbb{N}} \sim \mathbb{R}$ [illegible] $\mathcal{P}(\mathbb{N}) \sim \mathbb{R}$.

$\mathbb{R}\times\mathbb{R}\sim\mathbb{R}$

- [illegible] $F:\{0,1\}^{\mathbb{N}}\times\{0,1\}^{\mathbb{N}}\to\{0,1\}^{\mathbb{N}}$ [illegible]" [illegible]": $F(\langle a,b\rangle)=c$ [illegible]

$$c(2n)=a(n),\qquad c(2n+1)=b(n)$$

[illegible] $c=a_0\,b_0\,a_1\,b_1\,a_2\,b_2\ldots$

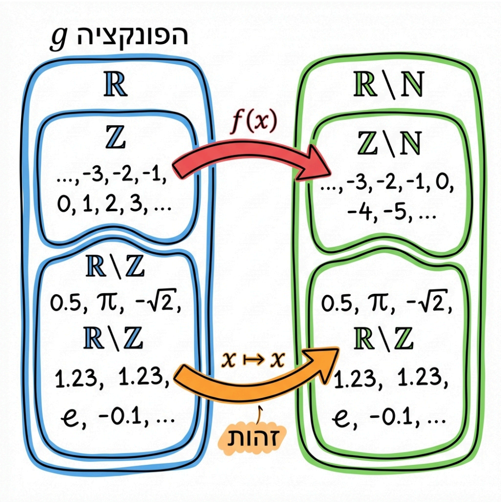
- F [illegible]": [illegible] $F(\langle a,b\rangle)=F(\langle a',b'\rangle)=c$, [illegible] n: $a(n)=c(2n)=a'(n)$ [illegible]-$b(n)=c(2n+1)=b'(n)$. [illegible] $a=a'$ [illegible]-$b=b'$.
- F [illegible]: [illegible] $c\in\{0,1\}^{\mathbb{N}}$ [illegible] $a(n)=c(2n)$ [illegible]-$b(n)=c(2n+1)$. [illegible] $F(\langle a,b\rangle)=c$.
- [illegible] $\{0,1\}^{\mathbb{N}}\times\{0,1\}^{\mathbb{N}}\sim\{0,1\}^{\mathbb{N}}$.
- [illegible] [illegible]-$\mathbb{R}\sim\{0,1\}^{\mathbb{N}}$, [illegible]:

$$\mathbb{R}\times\mathbb{R}\sim\{0,1\}^{\mathbb{N}}\times\{0,1\}^{\mathbb{N}}\sim\{0,1\}^{\mathbb{N}}\sim\mathbb{R}$$

[illegible]: [illegible] $\aleph_0$

- [illegible] A [illegible].
- [illegible] $a_0, a_1, a_2, \ldots$ [illegible]:
  - $a_0 \in A$.
  - [illegible] $n > 0$, [illegible] a_n [illegible] $A \setminus \{a_i \mid i < n\}$.
- [illegible], [illegible] A [illegible] n [illegible], [illegible] A [illegible].
- [illegible] a_n [illegible].
- [illegible] $\{a_n \mid n \in \mathbb{N}\}$ [illegible] A, [illegible] $\aleph_0$.

[illegible]: $\mathbb{R} \setminus \mathbb{N} \sim \mathbb{R}$


הפונקציה g
$\mathbb{R}$
$\mathbb{R}\setminus\mathbb{N}$
$f(x)$
$\mathbb{Z}$
…, -3, -2, -1,
0, 1, 2, 3, …
$\mathbb{Z}\setminus\mathbb{N}$
…, -3, -2, -1, 0,
-4, -5, …
$\mathbb{R}\setminus\mathbb{Z}$
0.5, π, -√2,
$\mathbb{R}\setminus\mathbb{Z}$
1.23, 1.23,
e, -0.1, …
$x \mapsto x$
$\mathbb{R}\setminus\mathbb{Z}$
0.5, π, -√2,
$\mathbb{R}\setminus\mathbb{Z}$
1.23, 1.23,
e, -0.1, …
זהות

- $\mathbb{Z} \setminus \mathbb{N} \sim \mathbb{N}$:
  - $\mathbb{Z} \setminus \mathbb{N}$ [illegible] $\mathbb{Z}$, [illegible].
  - [illegible].
- [illegible] $f : \mathbb{Z} \to \mathbb{Z} \setminus \mathbb{N}$ [illegible].
- [illegible] $g : \mathbb{R} \to \mathbb{R} \setminus \mathbb{N}$ [illegible]:

f(x) & x \in \mathbb{Z} \\ x & x \in \mathbb{R} \setminus \mathbb{Z} \end{cases}

- [illegible].

[illegible] [illegible]: $\mathcal{P}(\mathcal{P}(\mathbb{N}\times\mathbb{N})) \not\sim \mathbb{R}$

- [illegible] [illegible] [illegible]-$\mathcal{P}(\mathcal{P}(\mathbb{N}\times\mathbb{N})) \sim \mathbb{R}$.
- [illegible] [illegible]-$\mathbb{R} \sim \mathcal{P}(\mathbb{N})$.
- [illegible], [illegible] [illegible], $\mathcal{P}(\mathcal{P}(\mathbb{N}\times\mathbb{N})) \sim \mathcal{P}(\mathbb{N})$.
- [illegible] [illegible] [illegible]-$\mathcal{P}(\mathbb{N}\times\mathbb{N}) \sim \mathcal{P}(\mathbb{N})$.
- [illegible] [illegible] [illegible]-$\mathcal{P}(\mathcal{P}(\mathbb{N}\times\mathbb{N})) \sim \mathcal{P}(\mathcal{P}(\mathbb{N}))$.
- [illegible] $\mathcal{P}(\mathcal{P}(\mathbb{N}))$ [illegible] [illegible] [illegible] [illegible] [illegible]-$\mathcal{P}(\mathbb{N})$.
- [illegible] [illegible], [illegible] $\mathcal{P}(\mathcal{P}(\mathbb{N}\times\mathbb{N})) \not\sim \mathbb{R}$.

[illegible] [illegible]: [illegible] [illegible] $(A^B)^C \sim A^{B\times C}$

- [illegible] [illegible] $F : (A^B)^C \to A^{B\times C}$.
- [illegible] $f \in (A^B)^C$. [illegible] [illegible] [illegible] $c \in C$ [illegible] $f(c) \in A^B$. [illegible] $h_c = f(c)$. [illegible] $h_c : B \to A$.
- [illegible] [illegible] $g = F(f)$ [illegible] [illegible] $\langle b, c\rangle \in B \times C$: $g(b,c) = h_c(b) = (f(c))(b)$
- [illegible] F-[illegible] [illegible] (Surjective):
  - [illegible] $g \in A^{B\times C}$. [illegible] $f \in (A^B)^C$ [illegible]-$F(f) = g$.
  - [illegible] [illegible] f [illegible] $c \in C$, [illegible] $f(c) : B \to A$ [illegible]: $(f(c))(b) = g(b,c)$.
  - [illegible] [illegible], [illegible] [illegible] (b,c) [illegible] $(F(f))(b,c) = (f(c))(b) = g(b,c)$, [illegible] $F(f) = g$.
- [illegible] F-[illegible]-[illegible]-[illegible] (Injective):
  - [illegible] $F(f_1) = F(f_2) = g$.
  - [illegible] $c \in C$ [illegible] $b \in B$, [illegible] $(f_1(c))(b) = g(b,c) = (f_2(c))(b)$.
  - [illegible] $f_1(c)$-[illegible] $f_2(c)$ [illegible] [illegible] c, [illegible] $f_1 = f_2$.

[illegible]: $(\{0,1\}^{\mathbb{N}})^{\mathbb{N}} \sim \mathbb{R}$

- [illegible] $(\{0,1\}^{\mathbb{N}})^{\mathbb{N}}$:

$$|(\{0,1\}^{\mathbb{N}})^{\mathbb{N}}| = (2^{\aleph_0})^{\aleph_0}$$

- [illegible] ($(a^b)^c = a^{b\cdot c}$):

$$(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0}$$

- [illegible] $\aleph_0 \cdot \aleph_0 = \aleph_0$.
- [illegible]:

$$2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0}$$

- [illegible] $\mathfrak{c} = 2^{\aleph_0}$ ([illegible]).
- [illegible]:

$$(\{0,1\}^{\mathbb{N}})^{\mathbb{N}} \sim \mathbb{R}$$